Linear Representation of Lie algebra and Lie groups

Def^{n} [Linear Rep^{r} of a Lie algebra].

Let $(L, [\cdot,\cdot])$ be a Lie algebra. Then a linear representation ρ of this Lie algebra is:

$$\rho : L \xrightarrow{\sim} \mathrm{End}(V)$$

for some (finite-dim) vector space V over same field (as L).

such that

$$\rho([\![a, b]\!]) = [\rho(a), \rho(b)] \quad \forall\, a, b \in L.$$

The particular vector space V is called the rep^{r} space.

⊛ $\dim V$ is known as the dim of rep^{r}.

Idea: now we can act a element of Lie algebra on some vector from V via $\rho(a)$.

⊛ End(V)

We already know that $\mathrm{End}(V)$ itself is a vector space over same field as of V.

define Commutator:

$$\forall \phi, \psi \in \mathrm{End}(V): [\phi, \psi] = \phi \circ \psi - \psi \circ \phi$$

- anti-symmetric
- Jacobi identity

$$\Rightarrow (\mathrm{End}(V), [\cdot,\cdot]) \text{ is a Lie algebra.}$$

$$\Rightarrow \rho \text{ is a Lie algebra homomorphism.}$$

$$\mathrm{End}(V) = \mathrm{Hom}(V, V) = \{\phi : L \xrightarrow{\sim} L\}.$$

Example:

(a) $L = \mathfrak{sl}(2, \mathbb{C})$.

$\{X_1, X_2, X_3\}$ → Basis.

$$[\![X_1, X_2]\!] = 2X_2$$
$$[\![X_1, X_3]\!] = -2X_3$$
$$[\![X_2, X_3]\!] = X_1$$

a representation of $\mathfrak{sl}(2, \mathbb{C})$ is provided by

$$\rho : \mathfrak{sl}(2, \mathbb{C}) \longrightarrow \mathrm{End}(\mathbb{C}^2).$$

$$\rho(X_1) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}; \quad \rho(X_2) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \quad \rho(X_3) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Now we need to check the Lie bracket compatibility.

$$[\rho(X_1), \rho(X_2)] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} = 2\rho(X_2) = \rho(2X_2)$$
$$= \rho([\![X_1, X_2]\!]).$$

After checking, we are done. As the Lie bracket & commutator both are multilinear as well as ρ is a linear map, we are done by checking it for basis elements.

$$\Rightarrow \mathrm{End}(\mathbb{C}^2) \supseteq \rho(\mathfrak{sl}(2, \mathbb{C})) = \left\{ \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \;\middle|\; \alpha + \delta = 0 \right\}$$

Corresponding algebra is a vector space of traceless 2×2 matrices.

⊛ We say $SL(2, \mathbb{C})$ is group of invertible 2×2 matrices with unit determinat and

(b) $L = \mathfrak{so}(3,\mathbb{R})$. $\longrightarrow \dim_{\mathbb{R}} \mathfrak{so}(3,\mathbb{R}) = 3$.

$[\![X_i, X_j]\!] = C^k{}_{ij} X_k$; we need $C^k{}_{ij}$ to get the Lie bracket info.

def^n Killing form $\rightarrow$ Killing metric

$K_{ab} = C^m{}_{an} C^n{}_{bm}$ $\longrightarrow$ using this pull down "k"

$\Rightarrow C_{kij} := K_{km} C^m{}_{ij}$

for $\mathfrak{so}(3,\mathbb{R})$, $C_{kij} = \varepsilon_{ijk} = \begin{cases} +1, & (ijk) \text{ even permutation of } (1\,2\,3) \\ -1, & \text{odd perm} \\ 0, & \text{else.} \end{cases}$

representation $\rho : \mathfrak{so}(3,\mathbb{R}) \xrightarrow{\sim} \mathrm{End}(\mathbb{R}^3)$.

$$\rho(X_1) = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} ; \quad \rho(X_2) = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} ; \quad \rho(X_3) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}.$$

This is the rep^n that we know as rotation matrices.

⊛ In QM, we know another set of rep^n namely spinor.

$\rho_{spin} : \mathfrak{so}(3,\mathbb{R}) \longrightarrow \mathrm{End}(\mathbb{C}^2)$.

$$\rho_{spin}(X_1) = \frac{-i}{2}\sigma_1 = \frac{1}{2}\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} ; \quad \rho_{spin}(X_2) = \frac{-i}{2}\sigma_2 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} ; \quad \rho_{spin}(X_3) = \frac{1}{2}\sigma_3 = \frac{1}{2}\begin{pmatrix} -i & 0 \\ 0 & +i \end{pmatrix}$$

(c) trivial rep^n.

There is always the trivial rep^n on any vector space as the rep^n space

$\rho_{trivial} : L \xrightarrow{\sim} \mathrm{End}(V)$.

$\rho(a) := 0_{\mathrm{End}(V)}$; $0 : V \longrightarrow \{0\}$.

$0 = \rho([\![a,b]\!])$; $[\underbrace{\rho(a)}_{0}, \underbrace{\rho(b)}_{0}] = 0$.

(d) Adjoint rep^n: does every Lie algebra have a non-trivial rep^n?

Recall $\mathrm{ad} : L \xrightarrow{\sim} \mathrm{End}(L)$

$\mathrm{ad}(a)(b) := [\![a,b]\!]$ } This is also a rep^n, and it is important one as all of Lie theory is dependent on this.

To have this to be non-trivial,

$\forall a \in L : \mathrm{ad}(a) \neq 0$

$\Rightarrow \mathrm{ad}(a)(b) \neq 0 \quad \forall b \in L$.

$\Rightarrow \forall a, b \in L : [\![a,b]\!] \neq 0$. } $\rightarrow$ non-abelian Lie algebra.

⊠ Now, let's understand the building block of rep^n.

Def^n [Reducible and Irreducible].

A rep^n $\rho : L \xrightarrow{\sim} \mathrm{End}(V)$ is called reducible if there exists a vector subspace $U \neq \{0\} \subset V$ s.t.

$\forall a \in L : \forall u \in U : \rho(a)u \in U$. $\rightarrow$ U is called invariant subspace.

in other words, rep^n map ρ restricts to

$\rho|_U : L \xrightarrow{\sim} \mathrm{End}(U)$.

Otherwise, $\rho : L \xrightarrow{\sim} \mathrm{End}(V)$ is irreducible.

Example:

$\rho_{vec} : SO(3,\mathbb{R}) \longrightarrow \mathbb{R}^3$; $\rho_{spin} : SO(3,\mathbb{R}) \longrightarrow End(\mathbb{C}^2)$.

construct another rep^r

$\rho : SO(3,\mathbb{R}) \longrightarrow End(\mathbb{R}^3 \oplus \mathbb{C}^2)$.

$$\rho(a) \longmapsto \left(\begin{array}{c|c} \rho_{vec}(a) & 0 \\ \hline 0 & \rho_{spin}(a) \end{array}\right)$$

But $U = \mathbb{R}^3$ and $U' = \mathbb{C}^2$ are invariant subspace for this rep^r. $\Rightarrow$ ρ is reducible.

Def^n [Faithful]

A rep^r $\rho : L \xrightarrow{\sim} End(V)$ is called <u>faithful</u> if ρ is injective.

<u>Claim</u>: For a Lie algebra, check irreducibility and faithfulness.

<u>Pf</u>: Let $U \neq \{0\} \subset L$ be a invariant vector space.

$$\Rightarrow \forall a \in L \; \forall u \in U : ad(a)(u) \in U$$
$$\Rightarrow [\![a,u]\!] \in U$$
$$\Rightarrow [\![L,U]\!] \subseteq U$$
$$\Rightarrow U \text{ is an non-trivial ideal of } L.$$

⊛ If L is simple, then such U can't exist
$\Rightarrow$ no non-trivial invariant subspace
$\Rightarrow$ ad rep^r is irreducible.

Let $a, b \in L$, $\rho(a) = \rho(b) \Rightarrow a = b$ for injectivity.

⊛ $ad(a) = ad(b) \Rightarrow \forall h \in L : ad(a)(h) = ad(b)(h)$
$\Rightarrow [\![a,h]\!] = [\![b,h]\!]$
$\Rightarrow [\![a-b, h]\!] = 0 \quad \forall h \in L.$

Def^n center of Lie algebra : $Z(L) = \{x \in L \mid \forall h \in L : [\![x,h]\!] = 0\}$ $\longrightarrow$ all element of center commutes with all the other elements of algebra.

⊛ If $Z(L)$ is trivial $\Rightarrow$ $a - b = 0$ $\Rightarrow$ $\boxed{a = b}$
$\Rightarrow$ ad rep^r is faithful.

⊛ $Z(L) = \{0\} \iff$ ad rep^r is faithful.

⊛ L is simple $\iff$ ad rep^r is irreducible.

Def^n [Casimir Operator]

Casimir op^r can be constructed for any given rep^r of any given Lie algebra.

(a) $\rho : L \xrightarrow{\sim} End(V)$

def^n Killing form on L

$K_\rho : L \times L \longrightarrow \mathbb{C}$

$K_\rho(a,b) := tr(\rho(a) \circ \rho(b))$

Observe, the Killing form, we used till now was defined using ad rep^n. ($\rho = ad$).

(b) def Casimir Operator

$$\left.\begin{array}{l} \Omega_\rho : V \xrightarrow{\sim} V \end{array}\right\} \Rightarrow \Omega_\rho \in \mathrm{End}(V).$$

$$\Omega_\rho := \sum_{i=1}^{\dim L} \rho(X_i) \circ \rho(\xi_i)$$

where $X_1, \dots, X_{\dim L}$ is any basis of L

and $\xi_1, \dots, \xi_{\dim L}$ is the other basis of L

such that

$$K_\rho(X_i, \xi_j) = \delta_{ij} = \begin{cases} 1, & i=j \\ 0, & \text{else} \end{cases}$$

Theorem:

$$[\Omega_\rho, \rho(a)] = 0 \quad \forall a \in L. \Rightarrow \text{"Casimir opr commutes with everything } \rho(L)\text{"}$$

Proof: